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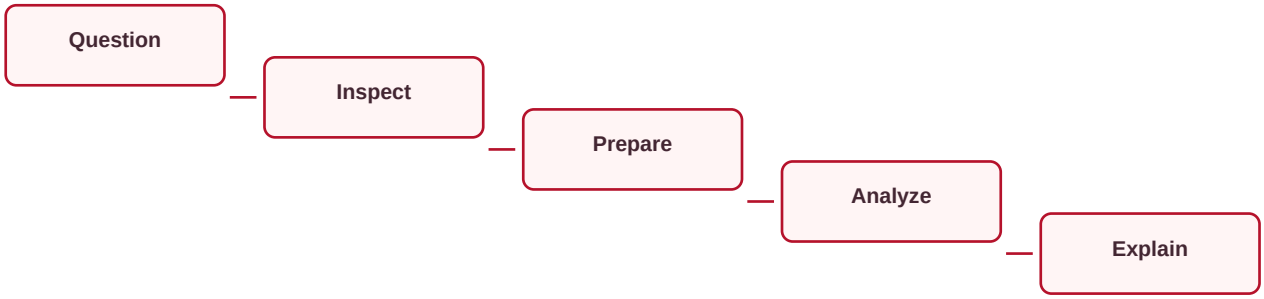
Data Science

DS101 | Suggested sequence · Semester 1 | Student-authored study guide

How to use these notes

A complete workflow for asking questions, preparing data, analyzing evidence, and communicating useful conclusions. Work through the units in order, reproduce each example, then answer the review prompts from memory. This is supplemental practice material, not an official MBU syllabus or assessment. Course sequencing is suggested and should be checked against your current approved syllabus.

A repeatable workflow



MODULE 01 · Questions, data, and the workflow

Translate a broad problem into a
window, and outcome before coll
model !’ communicate !’ monitor.

Worked example

```
For a campus travel survey, define one row as one respondent and ask: “What share of
respondents use public transport at least three days each week?” State how missing answers are
handled.
```

Check your understanding

- What is an observation in a dataset?
- Why should the question be written before choosing a model?

MODULE 02 · Data quality and exploratory analysis

Inspect dimensions, types, ranges, duplicates, missingness, and category frequencies. Use summaries and plots to find anomalies; do not silently discard inconvenient observations. Keep a reproducible record of every transformation.

Worked example

```
Compare a numeric column’s median and interquartile range before and after filtering
impossible negative durations. Investigate the original records before correcting them.
```

Check your understanding

- How can missingness itself be informative?
- When is a median more robust than a mean?

MODULE 03 · Modeling, validation, and communication

Separate training and evaluation data before fitting. Choose metrics that match the cost of errors, establish a simple baseline, check leakage and subgroup performance, and report limitations. A useful result explains both evidence and uncertainty.

Worked example

```
For rare-event classification, compare recall and precision at a stated threshold; accuracy
alone can hide a model that predicts the majority class every time.
```

Check your understanding

- What is data leakage?
- Why compare a model with a baseline?

Revision checklist

- %¡ Can I explain the main idea without copying the notes?
- %¡ Can I reproduce the example and state its assumptions?
- %¡ Can I test an edge case and describe what should happen?
- %¡ Have I recorded a limitation and checked a primary reference?

Further reading

Use the official documentation and open references linked in the portal. Check the publication/version and license at the source before redistribution. Cite the source and access date in coursework.